The Similarity Bistro

Assessment Menu · choose your meal and show what you know

Name: _____

Date: _____

How to order

1. Order **4 Appetizers** (5 points each), **4 Entrées** (10 points each) and **2 Chef's Specials** (20 points each). The most you can earn is **100 points**.
2. Your 10 dishes must include **at least one of each skill**: SCALE, CRITERIA, RATIOS and APPLY (the label next to each dish).
3. Mark your choices on the order form below **and** on the dishes themselves ("My pick"). Only marked dishes are graded, and you cannot swap a dish after turning it in.
4. Show your work in the space for each dish. Points are given for correct steps as well as correct answers.
5. Appetizers and Entrées are on pages 2 and 3. The Chef's Specials are on page 4. Look at every page before you choose!

Appetizers

5 pts each · choose 4

- ☐ **A1** SCALE
- ☐ **A2** SCALE
- ☐ **A3** CRITERIA
- ☐ **A4** CRITERIA
- ☐ **A5** RATIOS
- ☐ **A6** APPLY

Entrées

10 pts each · choose 4

- ☐ **E1** SCALE
- ☐ **E2** CRITERIA
- ☐ **E3** RATIOS
- ☐ **E4** APPLY
- ☐ **E5** APPLY
- ☐ **E6** RATIOS

Chef's Specials

20 pts each · choose 2

- ☐ **S1** SCALE
- ☐ **S2** CRITERIA
- ☐ **S3** RATIOS
- ☐ **S4** APPLY

My skill check (circle each skill you have covered): SCALE · CRITERIA · RATIOS · APPLY

Course	Dishes	Points each	Possible	Score (teacher)
Appetizers	4	5	20	
Entrées	4	10	40	
Chef's Specials	2	20	40	
Total			**100**	

Appetizers

5 points each · choose 4 · Warm-ups

Name: _____

A1 SCALE ☐ My pick · 5 pts

Rectangle A is 4 by 6. Rectangle B is similar to A, and its shorter side is 10. Find the scale factor from A to B and the longer side of B.

A2 SCALE ☐ My pick · 5 pts

△ABC ~ △DEF with AB = 6, DE = 9, and BC = 10. Write a proportion and find EF.

A3 CRITERIA ☐ My pick · 5 pts

Triangle 1 has angles 50° and 60°. Triangle 2 has angles 60° and 70°. Are the triangles similar? Show the third angles and name the criterion.

A4 CRITERIA ☐ My pick · 5 pts

△ABC has AB = 4, AC = 6, ∠A = 50°. △DEF has DE = 10, DF = 15, ∠D = 50°. Are they similar? Show the ratios and name the criterion.

A5 RATIOS ☐ My pick · 5 pts

Two similar triangles have a scale factor of 3 : 5. The smaller one has perimeter 18. Find the perimeter of the larger one.

A6 APPLY ☐ My pick · 5 pts

A 2 m stick casts a 3 m shadow. At the same moment a tree casts a 12 m shadow. Write a proportion and find the tree’s height.

Entrées

10 points each · choose 4 · Multi-step problems

Name: _____

E1 SCALE ☐ My pick · 10 pts

△ABC ~ △DEF. AB = 8, DE = 12, BC = x + 2, and EF = 3x − 3. Find x, BC, and EF.

E2 CRITERIA ☐ My pick · 10 pts

△ABC has sides 6, 8, 10. △DEF has sides 9, 12, 15. △GHI has sides 9, 12, 16. Is △ABC similar to △DEF? To △GHI? Show the ratios and name the criterion.

E3 RATIOS ☐ My pick · 10 pts

Two similar triangles have areas 20 and 45. Find the ratio of their corresponding sides. If the smaller triangle has perimeter 14, find the perimeter of the larger.

E4 APPLY ☐ My pick · 10 pts

In △ABC, D is on AB and E is on AC with DE ∥ BC. AD = 6, DB = 4, and AE = 9. Find EC. If DE = 12, find BC.

E5 APPLY ☐ My pick · 10 pts

A mirror lies flat on the ground 6 m from the base of a flagpole. Maya stands 1.5 m past the mirror, on the same line, and just sees the top of the pole in it. Her eyes are 1.6 m high. How tall is the pole?

E6 RATIOS ☐ My pick · 10 pts

Two similar triangles have sides in the ratio 2 : 3. The smaller has area 12. Dan finds the larger area as $12 \times \frac{3}{2} = 18$. Find his mistake, explain it, and give the correct area.

Chef’s Specials

20 points each · choose 2 · Multi-part challenges

Name: _____

S1 SCALE ☐ My pick · 20 pts

Quadrilaterals ABCD ~ WXYZ (in that order). AB = 6, WX = 9, BC = 2x + 2, XY = 4x − 1. (a) Find x and the scale factor from ABCD to WXYZ. (b) WXYZ has perimeter 60; find the perimeter of ABCD. (c) ABCD has area 48; find the area of WXYZ.

S2 CRITERIA ☐ My pick · 20 pts

In △ABC, D is on AB and E is on AC with AD = 4, AB = 10, AE = 6, AC = 15. (a) Show △ADE ~ △ABC and name the criterion. (b) Explain why DE ∥ BC. (c) If BC = 20, find DE.

S3 RATIOS ☐ My pick · 20 pts

Two similar triangular sails: the larger has area 144 ft² and perimeter 48 ft; the smaller has perimeter 32 ft. (a) Find the scale factor from large to small. (b) Find the smaller area. (c) A side of the smaller is 8 ft; find the matching side of the larger. (d) Why does area change by the square?

S4 APPLY ☐ My pick · 20 pts

A lamp is H ft above the ground on a vertical pole. Nia, 6 ft tall, stands 20 ft from the base of the pole and her shadow is 10 ft long. (a) Find H and explain why the triangles are similar. (b) She walks to 30 ft from the pole. How long is her shadow now?